

Processando config 0, seed 0, teste: Rabbit  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Rabbit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

smultin\_MultinomialBitsOver test:

-----  
N = 2, n = 8388608, r = 0, s = 32, L = 38,  
Sparse = TRUE

Number of bits = n = 8388608  
Number of cells =  $2^L$  = 274877906944  
Expected number per cell =  $1 / 32768$   
EColl =  $n^2 / (2k)$  = 128  
Hashing = TRUE

Collision test  
CollisionOver: density =  $n / k = 1 / 32768.00$   
Expected number of collisions =  $\mu = 128.00$

-----  
Results of CollisionOver test:

POISSON approximation :  
Expected number of collisions =  $N \cdot \mu$  : 256.00  
Observed number of collisions : 588  
p-value of test :  $2.0e-70$  \*\*\*\*\*

-----  
Total number of cells containing j balls

|       |   |              |
|-------|---|--------------|
| j = 0 | : | 549739037260 |
| j = 1 | : | 16776040     |
| j = 2 | : | 588          |
| j = 3 | : | 0            |
| j = 4 | : | 0            |
| j = 5 | : | 0            |

-----  
CPU time used : 00:00:04.62

Generator state:  
16777216 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 27.00  
p-value of test : 4.8e-7 \*\*\*\*\*

-----  
CPU time used : 00:00:00.05

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 17.00  
p-value of test : 1.8e-12 \*\*\*\*\*

-----  
CPU time used : 00:00:00.05

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : -0.37  
p-value of test : 0.64

-----  
CPU time used : 00:00:00.00

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 5.50  
p-value of test : 0.60

-----  
Normal statistic for number of jumps : -85.13  
p-value of test : 1 - eps1 \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : -0.48  
p-value of test : 0.68

-----  
CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : 0.47  
p-value of test : 0.32

-----

CPU time used : 00:00:00.23

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.039  
p-value of test : 0.21

Kolmogorov-Smirnov- statistic = D- : 0.026  
p-value of test : 0.49

Anderson-Darling statistic = A2 : 0.77  
p-value of test : 0.50

-----

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 3.09  
p-value of test : 0.38

-----  
Global longest run of 1 : 23.00  
p-value of test : 0.50

-----  
CPU time used : 00:00:00.05

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 105.11  
p-value of test : 0.20

-----  
CPU time used : 00:00:00.01

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 26.24  
p-value of test : 0.24  
-----

CPU time used : 00:00:00.00

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 5.01e-3  
p-value of test : 0.50  
-----

CPU time used : 00:00:00.00

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : 1.96  
p-value of test : 0.02  
-----

CPU time used : 00:00:00.00

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----  
Normal statistic : -1.48  
p-value of test : 0.93

-----  
CPU time used : 00:00:00.00

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 177  
Chi-square statistic : 181.61  
p-value of test : 0.39

-----  
CPU time used : 00:00:00.00

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 3603.04  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 2089.11  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          : -0.034
p-value of test          : 0.51
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config0_seed0.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          : -0.36
p-value of test          : 0.64
-----
```

```
CPU time used            : 00:00:00.00
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config0_seed0.bin
```

```
sstring_Run test:
```

```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355443
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 28.56
p-value of test : 0.81
-----
```

```
-----
Total number of bits: 13428096
```

```
Normal statistic for number of bits : 1.22
p-value of test : 0.11
```

-----  
CPU time used : 00:00:00.04

Generator state:  
299705184 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 2.64  
p-value of test : 0.45

-----  
CPU time used : 00:00:00.07

Generator state:  
316482400 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 5.83  
p-value of test : 0.05

-----  
CPU time used : 00:00:00.08

Generator state:  
333173600 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 42      |       |
| ChiSquare statistic          | : | 2888.55 |       |
| p-value of test              | : | eps     | ***** |

-----  
Test on the values of the Statistic M

|                              |   |         |       |
|------------------------------|---|---------|-------|
| Number of degrees of freedom | : | 42      |       |
| ChiSquare statistic          | : | 1189.43 |       |
| p-value of test              | : | eps     | ***** |

-----  
Test on the values of the Statistic J

|                              |   |        |       |
|------------------------------|---|--------|-------|
| Number of degrees of freedom | : | 64     |       |
| ChiSquare statistic          | : | 273.01 |       |
| p-value of test              | : | eps    | ***** |

-----  
Test on the values of the Statistic R

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 36    |  |
| ChiSquare statistic          | : | 45.19 |  |
| p-value of test              | : | 0.14  |  |

-----  
Test on the values of the Statistic C

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 22    |  |
| ChiSquare statistic          | : | 39.44 |  |
| p-value of test              | : | 0.01  |  |

-----  
CPU time used : 00:00:00.06

Generator state:  
349950816 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |        |       |
|------------------------------|---|--------|-------|
| Number of degrees of freedom | : | 94     |       |
| ChiSquare statistic          | : | 191.46 |       |
| p-value of test              | : | 1.2e-8 | ***** |

-----  
Test on the values of the Statistic M

|                              |   |        |       |
|------------------------------|---|--------|-------|
| Number of degrees of freedom | : | 94     |       |
| ChiSquare statistic          | : | 200.30 |       |
| p-value of test              | : | 1.1e-9 | ***** |

-----  
Test on the values of the Statistic J

|                              |   |        |  |
|------------------------------|---|--------|--|
| Number of degrees of freedom | : | 512    |  |
| ChiSquare statistic          | : | 526.84 |  |
| p-value of test              | : | 0.32   |  |

-----  
Test on the values of the Statistic R

|                              |   |        |  |
|------------------------------|---|--------|--|
| Number of degrees of freedom | : | 90     |  |
| ChiSquare statistic          | : | 106.07 |  |
| p-value of test              | : | 0.12   |  |

-----  
Test on the values of the Statistic C

|                              |   |       |  |
|------------------------------|---|-------|--|
| Number of degrees of freedom | : | 50    |  |
| ChiSquare statistic          | : | 58.05 |  |
| p-value of test              | : | 0.20  |  |

-----  
CPU time used : 00:00:00.04

Generator state:

366728032 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config0\_seed0.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 143.96 |
| p-value of test              | : | 0.19   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 156.61 |
| p-value of test              | : | 0.06   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 176.53 |
| p-value of test              | : | 0.19   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 103.48 |
| p-value of test              | : | 0.96   |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 89    |
| ChiSquare statistic          | : | 92.29 |
| p-value of test              | : | 0.38  |

-----  
CPU time used : 00:00:00.04

Generator state:

383504832 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:06.11  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

|    | Test                      | p-value  |
|----|---------------------------|----------|
| 1  | MultinomialBitsOver       | 2.0e-70  |
| 2  | ClosePairsBitMatch, t = 2 | 4.8e-7   |
| 3  | ClosePairsBitMatch, t = 4 | 1.8e-12  |
| 5  | LinearComp                | 1 - eps1 |
| 16 | HammingIndep, L = 32      | eps      |
| 17 | HammingIndep, L = 64      | eps      |
| 24 | RandomWalk1 H             | eps      |
| 24 | RandomWalk1 M             | eps      |
| 24 | RandomWalk1 J             | eps      |
| 25 | RandomWalk1 H (L = 1024)  | 1.2e-8   |
| 25 | RandomWalk1 M (L = 1024)  | 1.1e-9   |

-----  
All other tests were passed